

Music Streaming Application Using Data Structures

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INTRODUCTION

- A music streaming application allows users to listen to songs online.
- Users can search, play, and download music.
- Users can create and manage playlists.
- The application recommends songs based on user preferences.
- Data structures help organize and manage data efficiently.

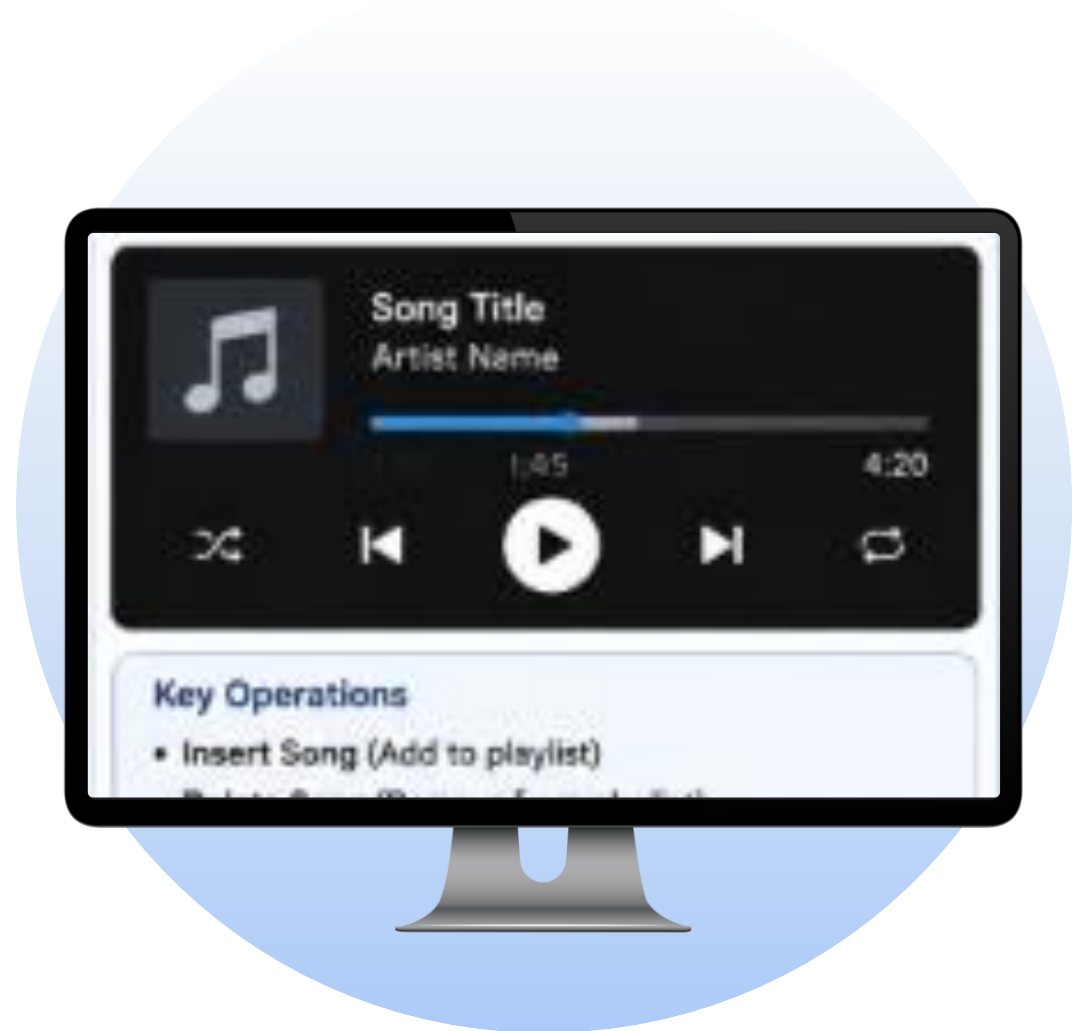
Operations

- **Insert:** Add new songs, users, and playlists.
- **Delete:** Remove songs or playlists.
- **Search:** Find songs by title, artist, or album.
- **Update:** Edit song details or playlists.
- **Traverse:** Display songs in playlists.
- **Sort:** Arrange songs by name, artist, or release date



Applications

- Music streaming platforms (Spotify, Apple Music, YouTube Music).
- Online radio services.
- Podcast streaming applications.
- Personalized music recommendation systems.
- Playlist management systems.
- Music library organization.



Conclusion

- Data structures improve the performance of music streaming applications.
- They enable quick searching and efficient storage.
- Different data structures serve different purposes.
- They provide a smooth and personalized listening experience.
- Data structures are essential for building modern music streaming platforms.

Dictionary Application

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Introduction

A dictionary application helps users find the meanings of words.

It provides synonyms, antonyms, pronunciation, and examples.

Users can search words quickly and efficiently.

It supports multiple languages and vocabulary learning.

Data structures help organize and retrieve words efficiently.



OPERATIONS

- Insert Word
- Search Word
- Delete Word
- Update Word
- Prefix Search
- Auto-complete Search

OPERATION	COMPLEXITY
Insert	$O(L)$
Search	$O(L)$
Delete	$O(L)$
Prefix search	$O(L)$

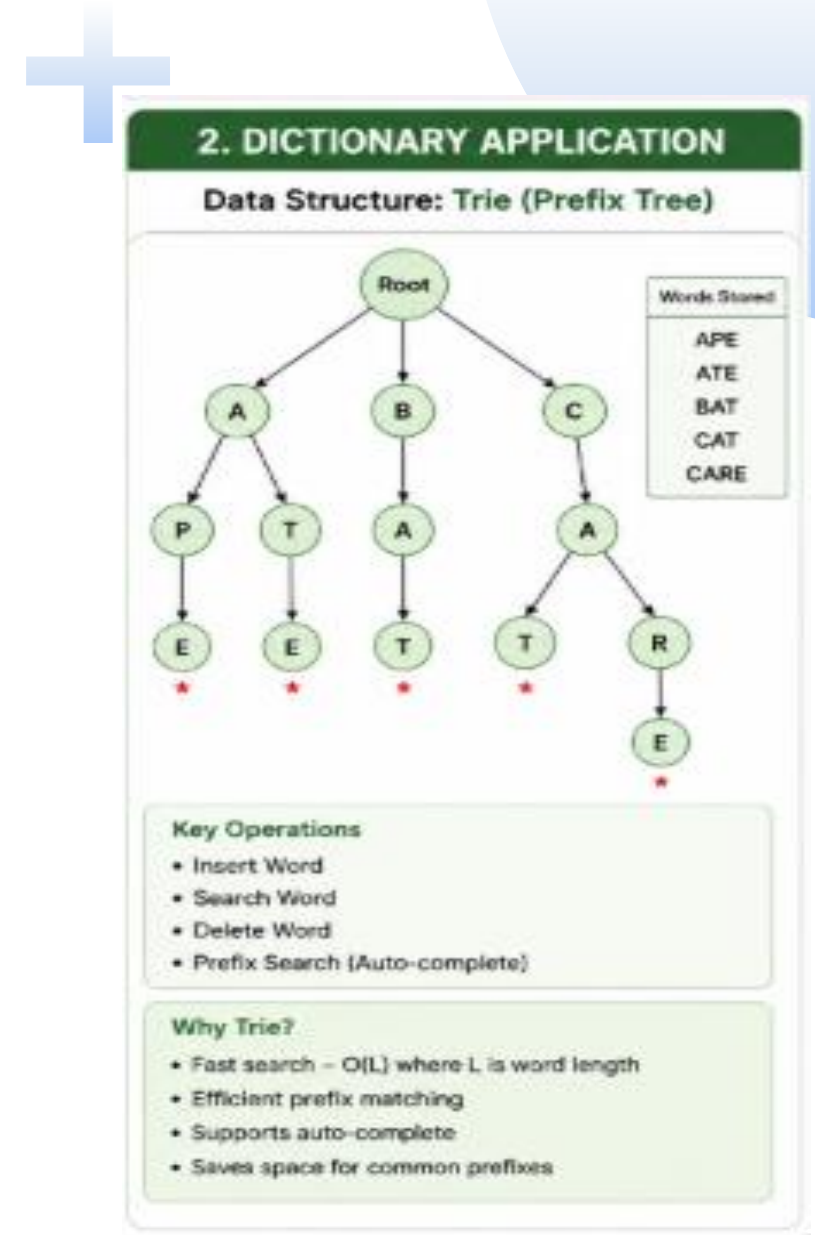
Applications

- Dictionary Applications
- Spell Checker
- Auto-complete Systems
- Google Search Suggestions
- Contact Search
- Search Engines
- IP Routing Tables



Conclusion

- ❖ Trie provides fast word searching.
- ❖ Supports prefix matching.
- ❖ Efficient for large dictionaries.
- ❖ Best choice for dictionary applications.



Task Scheduling System



Introduction

- **Task Scheduling – Priority Queue**
- A scheduler executes tasks based on priority.
- High-priority tasks are processed first.
- Used in operating systems and cloud computing.
- **Recommended Data Structure:** Priority Queue.
- Implemented using a Binary Heap.



Operations

- Insert Task
- Delete Highest Priority Task
- Peek Highest Priority Task
- Display Queue
- Update Priority

Operations	Complexity
Insert	$O(\log n)$
Delete	$O(\log n)$
Peek	$O(1)$
Build Heap +	$O(n)$

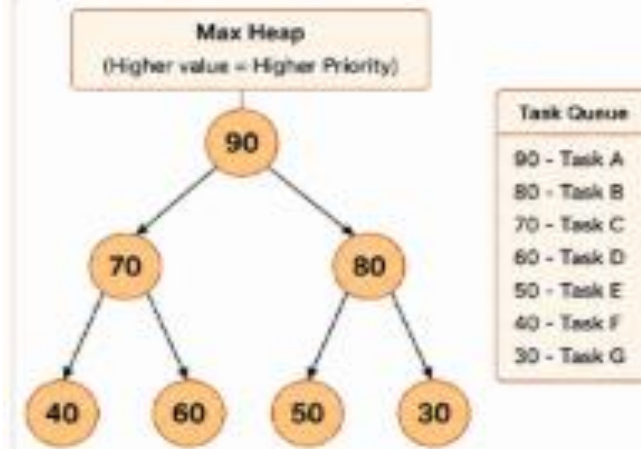


Applications

- CPU Scheduling
- Printer Scheduling
- Hospital Emergency Systems
- Network Packet Scheduling
- Airline Reservation Systems
- Cloud Computing
- Event Management Systems

3. TASK SCHEDULING SYSTEM

Data Structure: Priority Queue (Heap)



Key Operations

- Insert Task
- Get Highest Priority Task (Extract Max)
- Peek Highest Priority Task
- Update Priority

Why Priority Queue (Heap)?

- Always returns highest priority task
- Insert and delete in $O(\log n)$
- Efficient for scheduling and real-time systems
- Widely used in OS, cloud, networking

Conclusion

- ❖ Priority Queue executes tasks based on priority.
- ❖ Ensures efficient scheduling.
- ❖ Reduces waiting time for important tasks.
- ❖ Best suited for task scheduling systems



Thank you

